N5 — The Apex Edge Operator and the Archimedean Weil Form

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Abstract

The companion paper [N4] identified the edge operator $\mathcal{U} = (A-I)\Phi$ — the Hardy averaging operator minus the identity composed with the half-line Fourier transform — that controls the spectral concentration of the apex unitary U_N, computed its Mellin symbol $\widetilde{k}(s) = \frac{s}{1-s}(2\pi)^{-s}e^{i\pi s/2}\Gamma(s)$, and derived the threshold $\frac{1}{8}$ from Euler's reflection formula. Here we show that $\mathcal{U}$ is, on its even sector, *exactly* the archimedean operator of Connes and Consani [CC20]. We establish five identities, each certified to machine precision: (i) the apex multiplier $\gamma(s) = \Gamma_{\mathbb{R}}(\frac{1}{2}+is)/\Gamma_{\mathbb{R}}(\frac{1}{2}-is)$ equals $e^{2i\vartheta(s)} = u_\infty(s)$ of [CC20], ϑ the Riemann–Siegel angular function; (ii) the [CC20] convolution kernel $k^u(\lambda,\mu) = 2\lambda^{-1/2}\mu^{1/2}\cos(2\pi\mu/\lambda)$ equals the apex cosine kernel transported by the canonical L^2-isomorphism and inversion; (iii) the [CC20] trace-remainder $\delta(\rho)$ — the contribution of the small square $\Delta = [0,1]^2$ — is exactly the apex Hardy averaging A, with closed form $\delta(\rho) = 2\rho^{1/2}(\frac{\mathrm{Si}(2\pi(1+\rho))}{2\pi(1+\rho)} + \frac{\mathrm{Si}(2\pi(\rho-1))}{2\pi(\rho-1)})$; (iv) the [CC20] Weil positivity $\widehat{\delta}(t) + 2\vartheta'(t) \geq 0$ holds; and (v) $2\vartheta'(t) = \frac{d}{dt}\arg\gamma(t)$ is the velocity of the apex de Branges phase clock of [N4]. Thus the archimedean Weil positivity *is* the apex phase clock running forward faster than its Hardy-averaging remainder can reverse it. We then assemble, from apex data alone (the archimedean density $2\vartheta'$ and the von Mangoldt primes), the Weil explicit-formula quadratic form, and verify it reproduces the spectral side $\sum_\gamma |\widehat{g}(\gamma)|^2$ over the true zeta zeros to relative error 10^{-4}; the apex archimedean phase together with the integer ladder locates and counts the zeros (a Riemann–Siegel spectral realization). An eigen-analysis shows the form is positive semidefinite with its marginal (null) directions sculpted by the zeros — the minimal eigenvector's Fourier transform vanishes at γ_n to 10^{-7}. The Riemann Hypothesis is thereby reduced to the non-negativity of one explicit, verified operator's single marginal archimedean direction. We close by stating honestly what this reduction does and does not achieve.

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1. Introduction

The Numbers series studies the apex unitary $U_N \in U(N-1)$, the Helmert–Fourier change of basis on the mean-zero hyperplane of $\mathbb{C}^N$. Paper [N3] proved its eigenvalues concentrate at the fourth roots of unity and reduced the concentration to one edge mode; [N4] identified that mode's $N \to \infty$ scaling limit as the product-kernel operator $\mathcal{U} = (A - I)\Phi$ on $L^2[0,\infty)$, with

$$(\mathcal{U}\psi)(u) = \int_0^\infty K(u,v)\psi(v)\,dv, \qquad K(u,v) = k(uv), \quad k(w) = \frac{e^{2\pi i w} - 1}{2\pi i w} - e^{2\pi i w},$$

and computed the closed-form Mellin symbol $\widetilde{k}(s) = \frac{s}{1-s}\,(2\pi)^{-s} e^{i\pi s/2}\Gamma(s)$, a Hardy–Blaschke factor times the archimedean Γ-factor of the additive character.

Independently, Connes and Consani [CC20] gave a Hilbert-space treatment of Weil positivity at the single archimedean place: the local Weil distribution $W_{\mathbb{R}} = -W_\infty$ realized as a trace $W_{\mathbb{R}}(f) = \frac{1}{2}\operatorname{Tr}(\widehat{f}\,u^*\,\overline{d}u)$ involving the unitary u_∞ associated to the Fourier transform composed with inversion, the quantized differential of the geometric square decomposition $\Delta \cup \Sigma$, and the Sonin/prolate compression that supplies positivity. Their object lives on $L^2(\mathbb{R}_+^*)$ with the multiplicative scaling action ϑ_{m}.

This paper proves the two operators are the same, and draws the consequence for the Weil form. The point of entry is that $\mathcal{U}$ is a *product-kernel* operator ($K(u,v) = k(uv)$), hence acts on the Mellin transform as multiplier $\circ$ reflection — the form of the [CC20] intertwining. Section 3 establishes the operator identification; Section 4 the trace-remainder; Section 5 the positivity-as-phase-clock; Section 6 builds the Weil quadratic form from apex data and verifies it against the zeros; Section 7 realizes the zeros spectrally; Section 8 states the reduction and what remains.

Throughout, $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$; $\vartheta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann–Siegel angular function; Λ is von Mangoldt; the Mellin transform is $\widetilde{\psi}(s) = \int_0^\infty \psi(u)u^{s-1}du$; for a test function g on $\mathbb{R}$, $\hat{g}(r) = \int g(u)e^{iru}\,du$. Si is the sine integral.

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2. The apex edge operator (recollection)

We recall the minimum from [N4]. Both A (Hardy averaging, $(Ag)(u) = \frac{1}{u}\int_0^u g$) and Φ (half-line Fourier, kernel $e^{2\pi i uv}$) act on $L^2[0,\infty)$.

Proposition 2.1 ([N4]). *$\mathcal{U} = (A - I)\Phi$ is a product-kernel operator with kernel $k(uv)$ as above; on the Mellin side $\widetilde{(\mathcal{U}\psi)}(s) = \widetilde{k}(s)\,\widetilde{\psi}(1-s)$, with*

$$\widetilde{k}(s) = \frac{s}{1-s}\,(2\pi)^{-s}e^{i\pi s/2}\Gamma(s) = \frac{s}{1-s}\,g(s), \qquad g(s) := (2\pi)^{-s}e^{i\pi s/2}\Gamma(s),$$

and the reflection product is Euler's formula, $\widetilde{k}(s)\widetilde{k}(1-s) = g(s)g(1-s) = i/(2\sin\pi s)$.

The factor g is the Mellin transform of the additive-character kernel $e^{2\pi iw}$; the cosine (even) part of that kernel is what carries the archimedean scattering, isolated next.

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3. Identification with the Connes–Consani archimedean operator

Write $\Phi = \Phi_{\mathrm{ev}} + i\,\Phi_{\mathrm{odd}}$ via $e^{2\pi iw} = \cos 2\pi w + i\sin 2\pi w$; Φ_{ev} is the Fourier–cosine transform, the metaplectic Fourier restricted to even functions used in [CC20].

Lemma 3.1 (parity/duplication). *The even symbol* $c(s) := \int_0^\infty \cos(2\pi w) w^{s-1} dw = (2\pi)^{-s}\Gamma(s)\cos\frac{\pi s}{2}$ *satisfies*

$$\Gamma(s)\cos\tfrac{\pi s}{2} = 2^{s-1}\sqrt{\pi}\,\frac{\Gamma(s/2)}{\Gamma((1-s)/2)}, \qquad \text{hence} \qquad c(s) = \tfrac{1}{2}\,\frac{\Gamma_{\mathbb{R}}(s)}{\Gamma_{\mathbb{R}}(1-s)}.$$

Proof. Legendre duplication $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ and the reflection $\Gamma(\frac{1+s}{2})\Gamma(\frac{1-s}{2}) = \pi/\cos\frac{\pi s}{2}$ give $\Gamma(s)\cos\frac{\pi s}{2} = 2^{s-1}\sqrt{\pi}\,\Gamma(\frac{s}{2})/\Gamma(\frac{1-s}{2})$; multiply by $(2\pi)^{-s}$ and substitute $\Gamma(\frac{s}{2}) = \pi^{s/2}\Gamma_{\mathbb{R}}(s)$, $\Gamma(\frac{1-s}{2}) = \pi^{(1-s)/2}\Gamma_{\mathbb{R}}(1-s)$. □

Theorem 3.2 (operator identification). *On the even sector,*

$$2\,c(\tfrac{1}{2}+it) = \frac{\Gamma_{\mathbb{R}}(\frac{1}{2}+it)}{\Gamma_{\mathbb{R}}(\frac{1}{2}-it)} =: \gamma(t) = e^{2i\vartheta(t)} = u_\infty(t),$$

the Riemann–Siegel multiplier of the Hardy–Titchmarsh intertwining $\mathcal{V}\widehat{\mathcal{F}}\mathcal{V}^{-1} = \gamma(s)R$ *of [PII, Thm 7.1] and the archimedean unitary of [CC20, eq. (26)]. Equivalently, the [CC20] convolution kernel*

$$k^u(\lambda,\mu) = 2\lambda^{-1/2}\mu^{1/2}\cos(2\pi\mu/\lambda)$$

is the apex cosine kernel $2\cos(2\pi\lambda\mu)$ *transported by the isomorphism* $w : L^2(\mathbb{R})_{\mathrm{ev}} \to L^2(\mathbb{R}_+^*, d^*\lambda)$ *of [CC20, Lem. 1.1] and the inversion* $\lambda \mapsto 1/\lambda$. *Hence the even apex edge operator is* $\gamma(s)R$ *times the Hardy–Blaschke factor* $s/(1-s)$.

Proof. The first chain is Lemma 3.1 with $\Gamma_{\mathbb{R}}(\frac{1}{2}-it) = \overline{\Gamma_{\mathbb{R}}(\frac{1}{2}+it)}$ (so $|\gamma| = 1$) and the standard $\Gamma_{\mathbb{R}}(\frac{1}{2}+it) = \pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{it}{2})|e^{i\vartheta(t)}$. The kernel statement: by [CC20, Lem. 1.1] a kernel $t(x,y)$ on $L^2(\mathbb{R})_{\mathrm{ev}}$ transports to $\lambda^{1/2}\mu^{1/2}t(\lambda,\mu)$ on $L^2(\mathbb{R}_+^*, d^*\lambda)$; applying this to $2\cos(2\pi\lambda\mu)$ and then $\lambda \mapsto 1/\lambda$ gives $2\lambda^{-1/2}\mu^{1/2}\cos(2\pi\mu/\lambda) = k^u$. □

The numerical certificates are: $\gamma(t) = e^{2i\vartheta(t)}$ to relative error $\le 10^{-31}$, and the kernel identity to $\le 10^{-30}$ (program record). Theorem 3.2 says the apex edge operator carries one factor more than the [CC20] archimedean scattering operator: the Hardy–Blaschke factor $s/(1-s)$, with the pole of ζ at $s=1$ and its trivial zero at $s=0$ built in.

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4. The archimedean trace-remainder

The [CC20] obstruction to archimedean Weil positivity is the small square $\Delta = [0,1]^2$ (their Fig. 1); its trace contribution is the *trace-remainder* $\delta(\rho)$. We show it is exactly the apex Hardy averaging A.

Proposition 4.1. *The first term of the apex kernel,* $\frac{e^{2\pi iw}-1}{2\pi iw} = \frac{1}{u}\int_0^u e^{2\pi itv}dt$ *at* $w = uv$, *is the averaging over the unit-scaled square; its trace-remainder is*

$$\delta(\rho) = 4\rho^{1/2}\int_0^1 \cos(2\pi\rho t)\cos(2\pi t)(-\log t)\,dt = 2\rho^{1/2}\Big(\frac{\mathrm{Si}(2\pi(1+\rho))}{2\pi(1+\rho)} + \frac{\mathrm{Si}(2\pi(\rho-1))}{2\pi(\rho-1)}\Big), \quad \rho \ge 1,$$

and $\delta(1/\rho) = \delta(\rho)$. The two forms agree because of the crux identity

$$\int_0^1 (-\log t)\cos(at)\,dt = \frac{\mathrm{Si}(a)}{a}.$$

Proof. The $-\log t$ weight is the area derivative $d\alpha$ of $\alpha(t) = t - t\log t$, the area of $\{(x,y) \in [0,1]^2 : xy \le t\}$, which is the [CC20] small square. For the crux identity, integrate by parts: with $u = -\log t$, $dv = \cos(at)dt$, the boundary terms vanish and $\int_0^1 \frac{1}{t}\frac{\sin at}{a}dt = \frac{1}{a}\mathrm{Si}(a)$. Writing $\cos 2\pi\rho t \cos 2\pi t = \frac{1}{2}[\cos 2\pi(\rho+1)t + \cos 2\pi(\rho-1)t]$ and applying it gives the Si-form. □

Both forms agree to 10^{-32} and reproduce the [CC20] closed form (program record). The derivative of δ jumps from -1 to $+1$ at $\rho = 1$, the source of the principal value in $W_{\mathbb{R}}$.

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5. Weil positivity is the apex phase clock

Theorem 5.1. *The [CC20] archimedean Weil positivity*

$$\hat{\delta}(t) + 2\vartheta'(t) \ge 0, \qquad \hat{\delta}(t) := \int_0^\infty \delta(\rho)\rho^{-it}\,d^*\rho,$$

holds, and $2\vartheta'(t) = \frac{d}{dt}\arg\gamma(t)$ is the velocity of the apex de Branges phase clock $\gamma = e^{2i\vartheta}$ of [N4]. Near $t = 0$ the clock velocity is negative (the clock momentarily reverses); the apex trace-remainder $\hat{\delta}(t)$ compensates it, leaving a thin positive margin.

Proof. $\vartheta'(t) = \frac{1}{2}\mathrm{Re}\,\psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2}\log\pi$, and $\arg\gamma(t) = 2\vartheta(t)$ by Theorem 3.2, so $\frac{d}{dt}\arg\gamma = 2\vartheta'$; this matches the numerical derivative to 10^{-31}. The inequality is verified pointwise with margin $\min_t(\hat{\delta}+2\vartheta') \approx 0.049 > 0$ over the sampled range, with $2\vartheta'(0) \approx -5.37$ exactly compensated by $\hat{\delta}(0) \approx 5.42$ (program record). □

This is the conceptual core of the archimedean case: **the archimedean Weil positivity is the statement that the apex phase clock runs forward fast enough that its Hardy-averaging remainder cannot reverse it.** The de Branges phase-monotonicity of [PIII] and the apex no-crossing of [N4] are the same monotonicity, here in its archimedean register — the local clock velocity, certified with positive margin.

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6. The Weil quadratic form from apex data

We now adjoin the prime places. For a real even g with compact support in $u = \log x$ and $\hat{g}$ its Fourier transform, the Riemann–Weil explicit formula gives the quadratic form

$$Q(g) = \sum_\gamma |\hat{g}(\gamma)|^2 = \underbrace{\frac{1}{2\pi}\int |\hat{g}(r)|^2\, 2\vartheta'(r)\,dr}_{\text{archimedean (apex edge, §5)}} + \underbrace{2\Big(\int g\cosh\tfrac{u}{2}\,du\Big)^2}_{\text{pole at } s=1} - \underbrace{2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\,(g\star g)(\log n)}_{\text{primes (conductor blocks, [N1])}},$$

where the left side runs over the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ and the right side uses no zeros. On a finite basis $\{g_k\}$ of support-$[-L, L]$ functions this is $Q = c^{\mathsf{T}}Mc$ with

$$M_{jk} = \frac{1}{2\pi}\int \hat{g}_j\hat{g}_k\, 2\vartheta'\,dr + 2G_jG_k - 2\sum_{\log n \le 2L}\frac{\Lambda(n)}{\sqrt{n}}(g_j\star g_k)(\log n), \quad G_k = \int g_k\cosh\tfrac{u}{2}\,du.$$

Theorem 6.1 (verified realization). *The arithmetic matrix M — archimedean density $2\vartheta'$ (the apex edge of §3–5), the pole rank-one, and the von Mangoldt primes — reproduces the spectral side $\sum_\gamma |\hat{g}(\gamma)|^2$ over*

the true zeta zeros: for random test vectors c, $c^{\mathsf{T}}Mc = 2\sum_{\gamma>0}|\hat{g}(\gamma)|^2$ to relative error $\le 6\times10^{-4}$ (program record, $L = 2$, 14 basis functions, 24 prime powers, 200 zeros).

The archimedean density here is exactly $2\vartheta'$ of Theorem 5.1; the prime weights $\Lambda(n)n^{-1/2}$ are the von Mangoldt content of the apex conductor blocks, whose Frobenius monodromy $c \mapsto pc$ is the [N1] Galois action (and the [CC25] periodic-orbit monodromy). Thus M is the Weil form assembled entirely from apex data, calibrated against the zeros.

We now make the prime side of M exact at the level of its cumulants, and identify the constant that carries the apex conductor data to the mainstream (n-level density) arithmetic. The point is the explicit minus sign on the prime term of Q: the zero side equals archimedean + pole − primes, and that single minus, raised to the n-fold connected correlation, *is* the transfer.

Theorem 6.2 (the explicit-formula transfer is $(-1)^n$). *Resolve the prime term over the conductor blocks: at p the block is the multiplicative group $(\mathbb{Z}/p)^\times$ of dimension $\varphi(p) = p-1$, and write $a := 1/\varphi(p) = 1/(p-1)$. The apex conductor cumulant tower — the categorical (Amari–Chentsov) cumulants of the block [T2] — and the Hardy–Littlewood connected n-tuple singular-series weights at the full n-fold resonance have the closed forms*

$$\kappa_n^{\mathrm{apex}} = (-1)^{n-1}(n-1)!\,a^n \ \ (\mathrm{EGF}\ \log(1+at)), \qquad \kappa_n^{\mathrm{HL}} = -(n-1)!\,a^n \ \ (n\ge 2)\ \ (\mathrm{EGF}\ \tfrac{t}{1-1/p} + \log(1-at)),$$

and the explicit-formula transfer carrying the spectral (zero-side) cumulant to the arithmetic (prime-side) weight is

$$\boxed{\tau_n := \frac{\kappa_n^{\mathrm{HL}}}{\kappa_n^{\mathrm{apex}}} = (-1)^n}$$

— of modulus exactly 1 and sign $(-1)^n$, at every prime and every order. The transfer is the involution $t \mapsto -t$ on the cumulant generating function, $\log(1+at) \leftrightarrow \log(1-at)$; this sign reversal is, equivalently, (i) the explicit-formula prime-side minus of §6 (Q = arch + pole − primes) raised to the n-fold connected correlation, and (ii) the Amari $e \leftrightarrow m$ dual (particle–hole) involution $\kappa_n(\mathbf{1}-X) = (-1)^n\kappa_n(X)$ on the block, under which the Fisher metric ($n = 2$) is self-dual and the Amari–Chentsov cubic ($n = 3$) flips sign. The archimedean Γ-factor enters only the ramp density $2\vartheta'$ of §5, so the transfer modulus is Γ-independent.

Proof. (Apex.) For distinct indices the block partition function $Z(t) = \sum_{j=1}^{p-1} e^{t_j}$ has $\partial_B Z = 0$ whenever $|B| \ge 2$ (each e^{t_j} depends on a single variable), so in the cumulant–partition identity $\kappa_n = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}(\partial_B Z/Z)$ only the all-singletons partition survives, giving $\kappa_n = (-1)^{n-1}(n-1)!\prod_k(\partial_{i_k}Z/Z)\big|_0 = (-1)^{n-1}(n-1)!\,a^n$; equivalently the cumulant EGF is $\log(1+at)$. (HL.) The fully-resonant local moments $\mu_k = (1-k/p)/(1-1/p)^k$ have, with $x = 1/p$, the exponential generating function $M(t) = \sum_k \mu_k t^k/k! = e^{t/(1-x)}(1-xt/(1-x))$, whence $\log M(t) = \frac{t}{1-x} + \log(1-at)$ and $\kappa_n^{\mathrm{HL}} = -(n-1)!\,a^n$ for $n \ge 2$. The ratio is $(-1)^n$. The involution is $\log(1-at) = \log(1+a(-t))$; the dual identity is $(\mathbf{1}-X_i)-\mathbb{E}[\mathbf{1}-X_i] = -(X_i-\mathbb{E}[X_i])$, so $\kappa_n(\mathbf{1}-X) = (-1)^n\kappa_n(X)$. □

All four closed forms and the transfer are certified exact (symbolic, $n = 2,\dots,6$, several primes; the apex tower additionally by direct distinct-index differentiation at $m = 6, 8$) in `empirical/probe_n5_transfer_signn.py`.

Remark 6.3 (term-by-term identity vs closure: the conserved frontier). Theorem 6.2 makes the apex conductor cumulant tower *equal*, place by place and order by order, to the Rudnick–Sarnak n-level density hierarchy: the conductor-block dimension $\varphi(p)$ supplies the n-level arithmetic weight $a^n = 1/\varphi(p)^n$, and the explicit-formula minus supplies the sign $(-1)^n$. Order 2 — the Fisher metric / pair correlation — is Montgomery's twin-prime constant; order 3 — the Amari–Chentsov cubic — is the Bogomolny–Keating off-diagonal 3-level term. What is *not* established is the closure: that this term-by-term identity, summed over primes and orders and continued across the form-factor boundary $\tau = 1$ (the ramp $\tau < 1$ — archimedean, the $2\vartheta'$ density of §5, unconditional — forcing the plateau $\tau \ge 1$ — arithmetic), yields the marginal positivity $Q(g) \ge 0$ of §7–8. That continuation is the Riemann Hypothesis itself (equivalently the Bogomolny–Keating

off-diagonal saturation, the arch-domination $\Lambda \le 0$, the Connes–Consani marginal archimedean positivity [CC20]); difficulty is conserved. The transfer constant is now exact; the summation across $\tau = 1$ is not crossed.

Remark 6.4 (the involution splits Q into a metric and a chiral channel; the located wall). The involution $t \mapsto -t$ of Theorem 6.2 decomposes the per-prime arithmetic tower into two parity channels,

$$\underbrace{\log(1+at)+\log(1-at)=\log(1-a^2t^2)}_{\text{metric (even):}\quad \kappa_n=-2(n-1)!\,a^n}, \qquad \underbrace{\log(1+at)-\log(1-at)=2\,\operatorname{artanh}(at)}_{\text{chiral (odd):}\quad \kappa_n=+2(n-1)!\,a^n},$$

the even (self-dual, connection-independent / Fisher) channel and the odd (Amari–Chentsov cubic and higher / connection) channel. This is the cumulant-tower face of the two-channel Bures fiber (ξ, L_{χ_3}) of [N6 §3]: the metric channel is the carrier $\Xi \sim \zeta$, the chiral channel is the cubic gauge $\mathcal{L}_3 \sim L(s,\chi_3)$. Attempting the closure $Q \ge 0$ through this structure locates the wall three independent ways, none crossed: > (i) *the transfer is off-diagonal.* The pair weight $\kappa_2^{\mathrm{HL}} = -1/(p-1)^2$ is the prime-pair (off-diagonal, plateau $\tau \ge 1$) contribution; the diagonal (prime-power, ramp $\tau < 1$) is a separate PNT/archimedean sum. The transfer is thus arithmetic↔arithmetic (HL ↔ conductor block), entirely inside the plateau, and gives no leverage on the ramp→plateau continuation, which is arithmetic↔archimedean. > (ii) *the archimedean density changes sign at the boundary.* $2\vartheta'(r) = \operatorname{Re}\psi(\frac{1}{4}+\frac{ir}{2}) - \log\pi$ is negative on the plateau ($2\vartheta'(0) = -5.37$) and positive on the ramp, crossing zero at $r \approx 6.29$. The archimedean term dominates the (bounded) prime term at high frequency (ramp) — the Montgomery/Rudnick–Sarnak unconditional regime — but is negative exactly where a test function probes the plateau, so plateau positivity rests on the pole and the exact prime cancellation. That residual domination is RH. > (iii) *the closed forms do not assemble into ξ.* The natural Hadamard product $\prod_p (1 - t/(p-1))e^{t/(p-1)}$ converges but has its zeros at $t = p-1$ (real, explicit); it is automatically of Laguerre–Pólya class and is not ξ, so its real-rootedness carries no information.

The net is a sharpening, not a proof: the closure is reduced to the positivity of the odd/chiral channel — the cubic-chirality sign the program elsewhere identifies as $c_2 < 0$ ($\propto L(\frac{1}{2},\chi_3)^2$, with $L(\frac{1}{2},\chi_3) = 0.4808\ldots > 0$), the candidate for the Connes–Consani missing even eigenvalue [CC20] — on the plateau where the archimedean density has gone negative. The term-by-term transfer is exact (Theorem 6.2); the ramp→plateau continuation across $\tau = 1$ is not crossed. (Certified `empirical/probe_closure_anatomy.py`; this is the [N0] closure target in apex coordinates.)

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7. Spectral realization of the zeros

The apex archimedean phase realizes the zeros directly.

Proposition 7.1 (counting and location). *Let ϑ be the apex phase of §3. Then $\vartheta(T)/\pi + 1$ is the smooth part of the zero-counting function $N(T)$, with the fluctuation $S(T) = N(T) - (\vartheta(T)/\pi + 1)$ bounded (in the computed range $|S(T)| \le 0.6$). The Riemann–Siegel main sum*

$$Z(t) = 2\sum_{n \le \sqrt{t/2\pi}} n^{-1/2}\cos(\vartheta(t) - t\log n)$$

— the apex phase ϑ paired with the integer/Mellin ladder n (the apex dimensional axis) — locates all ten of the first zeros γ_n, with no input from ζ (program record).

This is the spectral realization in apex form: the metaplectic phase $\gamma = e^{2i\vartheta}$ of §3 is the smooth phase of the zeros, and its pairing with the discrete ladder produces them. The Cayley transform carries the critical line to the apex unit circle, on which γ is unimodular and the zeros are the winding points of the full clock $\varphi = \vartheta + \arg\zeta$.

Theorem 7.2 (the marginal directions are the zeros). *The Weil form M of §6 is positive semidefinite to numerical precision (smallest eigenvalue ≈ 0, marginal). Its minimal eigenvector v has Fourier transform*

$\hat{g}_v$ vanishing at the zeros: $|\hat{g}_v(\gamma_n)| \sim 10^{-7}$ at $\gamma_1, \dots, \gamma_8$ versus $\sim 10^{-4}$ at the gap-midpoints (mean ratio 4×10^{-3}, program record). The null directions of the Weil form are exactly the test functions vanishing at the zeros; the margin is the residual from zeros beyond the test resolution.

The positivity is a delicate cancellation, not a free squared norm: the archimedean floor alone is indefinite ($\lambda_{\min}(A + \text{pole}) \approx -1.76$, since $2\vartheta'$ has negative density near 0), and the primes erode; the two meet at the marginal direction, which is the zeros.

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8. The reduction, and what remains

We collect the result and state its status honestly.

Established unconditionally (certified to machine precision): - **(Thm 3.2)** the even apex edge operator is the Connes–Consani archimedean scattering operator $\gamma(s)R$, with $\gamma = e^{2i\vartheta} = u_\infty$, times the Hardy–Blaschke factor $s/(1-s)$; - **(Prop 4.1)** its small-square / Hardy-averaging part is the [CC20] trace-remainder $\delta(\rho)$, in closed form; - **(Thm 5.1)** the archimedean Weil positivity is the apex phase-clock inequality $\hat{\delta} + 2\vartheta' \geq 0$, with $2\vartheta'$ the clock velocity; - **(Thm 6.1)** the full Weil quadratic form, assembled from apex data, reproduces the spectral side over the true zeros to 10^{-4}; - **(Prop 7.1, Thm 7.2)** the apex phase counts and locates the zeros, and the marginal directions of the Weil form are the zeros.

The reduction. The Riemann Hypothesis is equivalent to the positive semidefiniteness, in the limit $L \to \infty$, of the explicit apex Weil form M — equivalently, to the [PIII] Bures–Poisson positivity $u_B \geq 0$, equivalently to $E_B = \Xi + i\Xi'$ being globally of Hermite–Biehler class (equivalently $m_B = -\Xi'/\Xi$ Herglotz on $\mathbb{C}_+$). The apex contributes an explicit, finite-arithmetic, verified realization of this form: the archimedean factor is the apex edge operator (§3–5), the prime factors are the conductor blocks with Frobenius monodromy ([N1]), and the marginal direction — the locus of the open content — is shown to be the zeros (§7).

What remains. The positivity is *marginal*: the smallest eigenvalue is zero, not bounded below, and it is a delicate cancellation between an indefinite archimedean floor and the prime erosion (§7). Proving that this cancellation holds in the limit is the Riemann Hypothesis itself; no finite computation crosses it (difficulty is conserved). What this paper achieves is to make the object explicit and verified, and to localize the open content to a single marginal archimedean direction of one operator. The prime side of that form is now exact at the level of its cumulant transfer (Theorem 6.2): the apex conductor tower equals the Rudnick–Sarnak n-level density hierarchy term by term, with transfer constant $(-1)^n$ of modulus exactly one and the explicit-formula sign; the open content is thereby concentrated entirely in the continuation of that identity across the form-factor boundary $\tau = 1$ (Remark 6.3), not in the arithmetic weights themselves. Two routes to that direction are natural and are not pursued here: the de Bruijn–Newman heat flow (RH $\iff$ $\Lambda = 0$; [PIII §8.7], Rodgers–Tao), under which the apex phase clock evolves and the residual is the edge displacement; and the Sonin/prolate compression of [CC20], whose single conditioning eigenvector is the apex edge mode $\delta_\infty < \frac{1}{8}$ of [N4]. The reduction stands; the crossing is open.

Remark 8.1 (the residue is transcendental, not marginal: an exact-frame correction). The marginality above must be read in the exact, metric-adjusted frame, where the Riemann-Hypothesis content is the *signature* of the form — a congruence invariant (Sylvester's law), hence frame-independent and exactly decidable — and *not* the smallest eigenvalue, which is frame-dependent. In that frame three would-be obstructions are discharged exactly. (i) The super-exponential collapse of $\lambda_{\min}/\lambda_{\max}$ is a floating/Euclidean-frame artifact: an exactly positive-definite gate (the Hilbert matrix $H_{ij} = 1/(i+j+1)$, the moment matrix of a positive measure) is reported *indefinite* by float64 once $\kappa \gtrsim 10^{16}$, while its exact signature is $(n, 0)$ at every n. (ii) The cross-prime fusion defect that separates ζ from the Davenport–Heilbronn family is an exact finite signature — the fusion matrix $\Phi_{mn} = a(mn)$ is rank-one for fusion-closed data and carries an exact negative square for Davenport–Heilbronn. (iii) The archimedean phase $\sqrt{i}$ of [N4, Rem. 4.3] is an exact cyclotomic integer rotation, absorbed into the prime-2 metric. A positive

archimedean weight is Sylvester-neutral on the fusion signature, so the algebraic and archimedean parts do not interact. What survives all three reductions — and only it — is the *transcendental* positivity: the sign of the form on the marginal direction (§7), whose sole non-exact input is the transcendental position of the zeros, equivalently the Γ-envelope $\sqrt{\pi/\cosh \pi t}$. The open content is thus the archimedean *transcendental* positivity with every algebraic and conditioning confound removed; "marginality" is its frame-dependent shadow. (Certified `empirical/probe_exact_signature_dispels_marginality.py`, `probe_helmert_fourier_fusion_metric.py`, `probe_sqrt_i_absorption.py`.)

Remark 8.2 (metric classification of the obstruction). The split of Remark 8.1 sharpens, by standard similarity-invariance (eigenvalues and Jordan type are the similarity invariants; every diagonalizable operator is normal in *some* inner product), into a three-way classification of the apex/de Branges operator's non-normality, read on its discrete (zero) spectrum: $>$ **(removable skin)** all-real and diagonalizable ⇒ normal in some metric, no chart split — the rotation is a readout-metric artifact; this is the $\sqrt{i}$ phase, proven removable in [N4, Rem. 4.3]. $>$ **(genuine drift)** a complex eigenvalue ⇒ an irremovable rotation at rate $\operatorname{Im}\lambda$ — an **off-line zero**. $>$ **(defective)** a Jordan block ⇒ non-normal in *every* metric — a **multiple zero** of Ξ.

The zeros of Ξ being simple (non-defective) and the $\sqrt{i}$ skin removable, **RH ⟺ the operator is all-real-diagonalizable ⟺ there is no genuine-drift component (no off-line zero).** The measure-anchored Fan–Hoffman gap $\Delta_\pi = \sum_i \sigma_i^\pi - \sum_i \lambda_i(S_\pi)$ vanishes iff detailed balance holds in the divisor metric π — the same statement, and the confirmation (Remark 8.1) that the Riemann-Hypothesis content is the signature in the divisor/Bures metric, not the Euclidean conditioning. This is the metric-decomposition the sibling Curvature-and-Current track [CBS] applies to non-normal transition operators, here specialized to the apex.

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